Diffusion Models Through the Lens of Tweedie's Formula

Empirical Bayes, a little SDE, and heavy tails

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Where we are going

The big picture

Tweedie's formula

Just enough SDE

Diffusion models

A baby/toy example

Tail behaviour: a Tauberian theorem

Summary

Before I begin

- I have only started exploring these topics (diffusion/ encoder / flow) a few weeks back and I am not an expert in these broad and deep area at all.
- I will present what I understand from a Bayesian statistician's viewpoint.
- If there are errors, please accept my apologies.

Sources

1. Handbook of Bayesian Deep Learning' by the BayesAI consortium. https://zenodo.org/records/22086450. (Chap. 50 is 'Diffusion models')
2. Score-based diffusion models via stochastic differential equations. Wenpin Tang and Hanyang Zhao. https://arxiv.org/abs/2402.07487
3. The Principles of Diffusion Models. Chieh-Hsin Lai et al. https://arxiv.org/abs/2510.21890v3
4. Consistent Diffusion Meets Tweedie, Daras et al. PyTorch implementation & paper: https://github.com/giannisdaras/ambient-tweedie

The big picture

Core Idea

- The core idea: a data sample can be progressively corrupted by adding noise until it becomes indistinguishable from pure randomness.
- This forward process moves different data samples into the same simple noise space, which is easy to sample from and serves as the starting point for generation.
- Along the way, it also creates a smooth sequence of intermediate noisy stages connecting data to noise.
- To generate, we draw a fresh noise sample from this space and gradually remove noise through a reverse-time (generation) process, eventually producing a new data sample.
- This reverse process does not recover one particular original example; instead, it starts from fresh noise and gradually turns it into realistic data.

Why ‘diffusion’?

- A diffusion model is a latent variable model that generates samples from the data distribution p_{data} by iteratively removing noise from a sample generated from a tractable prior distribution. To train such a denoiser, one needs to obtain noisy samples with varying degrees of noise.
- From the perspective of non-equilibrium thermodynamics, the noisy samples can be viewed as samples from an evolving system that moves from a low-entropy state towards a high-entropy state, for example, the diffusion of a gas in a container starting from some initial state.
- Recent successes: DALL-E 2 and Stable Diffusion, and the text-to-video generators Sora, Make-A-Video and Veo.

What a diffusion model does

A diffusion model is a forward–backward procedure.

Forward (deconstruct)

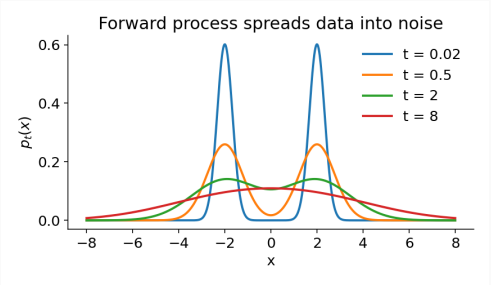
Start from data $X_0 \sim p_{\text{data}}$ and gradually add noise until the signal is gone: $X_T \approx$ pure noise.

Backward (construct)

Start from noise X_T and reverse the process to produce a fresh sample $X_0 \approx p_{\text{data}}$.

The forward part is trivial. The key part is the reverse.

"The answer hinges on two pillars: time reversal of (Markov) diffusion processes to set the form of the backward process, and score matching to learn this process."



forward process: data → noise

My main message

Connection:

A diffusion model applies **Tweedie's formula at every noise level**.

Outline:

1. **Tweedie's formula**: posterior mean from the marginal alone (empirical Bayes).
2. **Just enough SDE**: how to read $\mathrm{d}X_t = f\,\mathrm{d}t + g\,\mathrm{d}B_t$.
3. **Diffusion**: the reverse process needs only the *score*, and the score *is* the Tweedie correction.
4. **An image example**: noise an image, reverse it, no neural network.
5. **Tails**: a Tauberian theorem: the tail of the data fixes the *shape* of the optimal denoiser.

Tweedie's formula

The normal-means problem

Each unknown mean is drawn from a prior F; we see one noisy draw each:

$$\mu_i \sim F, \qquad Y_i \mid \mu_i \sim \mathcal{N}(\mu_i, \sigma^2), \qquad \sigma^2 \text{ known}.$$

Write the marginal density of an observation:

$$m(y) = \int \varphi_\sigma(y - \mu)\, \mathrm{d}F(\mu), \qquad \varphi_\sigma = \mathcal{N}(0, \sigma^2) \text{ density}.$$

Goal. The posterior mean $\mathbb{E}(\mu \mid Y = y)$: the Bayes rule under squared-error loss.

We assume that the Y_i's are exchangeable, i.e., invariant under permutation of the indices, which implies conditional independence and identical nature by de Finetti's theorem.

For Gaussian $f(\cdot)$, this problem is known as the normal means, or Gaussian sequence, or the Gaussian compound decision problem.

Empirical Bayes

- Compound decision problem [Robbins, 1951]: the estimates are allowed to depend on all observations $y = (y_1, \ldots, y_n)$, under the compound risk $\mathbb{E}_\mu\{\ell(\delta, \mu)\}$.
- The compound decision problem and Stein's shrinkage phenomenon [Stein, 1956] show that by allowing individual decisions/estimates to depend on the entire sequence, substantial reduction of total loss can be achieved ("borrowing strength from others").
- The empirical Bayes (EB) methodology, introduced by Robbins [1956], provides a statistical procedure where the goal of approximating the ideal or oracle Bayes rule is nearly achieved without specifying a prior [Zhang, 2003, Efron, 2011].
- A key appeal here is *observability* via Tweedie's formula, which depends only on the marginal $m(y)$, estimable from data without committing to a prior.

Two strategies for empirical Bayes: f-modeling vs. g-modeling

As emphasized by Efron [2014], empirical Bayes admits two complementary strategies:

- *f-modeling* models the marginal density $m(\cdot)$ on the observation (y) scale: the density of the data we actually see.
- *g-modeling* models the prior F on the parameter (μ) scale: the latent distribution the μ_i are drawn from.
- f-modeling has been relied upon heavily in applications, especially following Robbins [1956].
- g-modeling has been predominant in the theoretical EB literature [Laird, 1978, Jiang and Zhang, 2009].
- The two differ in what must be estimated: f-modeling estimates the observable marginal and reads off the Bayes rule by differentiating $\log m$ (Tweedie); g-modeling estimates the prior and must *deconvolve* the noise.

Tweedie's formula

Robbins (1956), attributed to M. C. K. Tweedie

$$\boxed{\mathbb{E}(\mu \mid Y = y) \;=\; y \;+\; \sigma^2 \,\frac{\mathrm{d}}{\mathrm{d}y} \log m(y).} \tag{T}$$

Read it as

$$\underbrace{\mathbb{E}(\mu \mid Y = y)}_{\text{Bayes estimate}} \;=\; \underbrace{y}_{\text{unbiased estimate}} \;+\; \underbrace{\sigma^2 (\log m)'(y)}_{\text{Bayes correction}}.$$

The correction pulls y toward where the data piles up. The prior F never appears: only the marginal m.

f-modeling vs. g-modeling

Recovering F from m is *deconvolution* (ill-posed). Tweedie sidesteps it entirely.

One-line derivation (Gaussian case, $\sigma^2 = 1$)

Let φ be the standard normal density, so $\varphi'(u) = -u\,\varphi(u)$.

$$m'(y) = \int \varphi'(y-\mu)\,\mathrm{d}F(\mu) = \int -(y-\mu)\,\varphi(y-\mu)\,\mathrm{d}F(\mu)$$
$$= -y\,m(y) + \int \mu\,\varphi(y-\mu)\,\mathrm{d}F(\mu) = -y\,m(y) + \mathbb{E}(\mu \mid y)\,m(y).$$

Divide by $m(y)$:

$$\frac{m'(y)}{m(y)} = -y + \mathbb{E}(\mu \mid y) \quad \Longrightarrow \quad \mathbb{E}(\mu \mid y) = y + (\log m)'(y).$$

That is all it is: differentiate the log-marginal.

Why it matters: empirical Bayes

The formula needs only m, and m is the density of the *observed* Y_i.

Empirical Bayes recipe

1. Estimate m (equivalently $\log m$) from all observations $y_1, \ldots, y_N$.
2. Differentiate: $\widehat{s}(y) = (\log \widehat{m})'(y)$.
3. Plug in: $\widehat{\mathbb{E}}(\mu_i \mid y_i) = y_i + \sigma^2\, \widehat{s}(y_i)$.

As $N \to \infty$ the correction is learned exactly: from the raw estimate at $N = 1$ to the true Bayes rule at $N = \infty$.

(Efron 2011 fits $\log m$ by Poisson regression on histogram counts: "Lindsey's method.")

The exponential-family version (Efron)

Tweedie is not special to the normal. For

$$y \mid \eta \sim e^{\eta y - \psi(\eta)} f_0(y), \qquad \eta \sim g,$$

set $\lambda(y) = \log\{m(y)/f_0(y)\}$. Then

$$\mathbb{E}(\eta \mid y) = \lambda'(y), \qquad \mathrm{Var}(\eta \mid y) = \lambda''(y).$$

- Differentiating the log-marginal returns the *posterior cumulants*.
- The normal case (T) is the instance with f_0 Gaussian.
- Higher derivatives give posterior skewness, etc.

James–Stein is Tweedie for a quadratic log-marginal

Take a Gaussian prior $\mu \sim \mathcal{N}(0, A)$. Then $m = \mathcal{N}(0, V)$ with $V = A + \sigma^2$, so (with $\sigma^2 = 1$) $(\log m)'(y) = -y/V$ and

$$\mathbb{E}(\mu \mid y) = \left(1 - \tfrac{1}{V}\right) y.$$

Replace the unknown $1/V$ by the unbiased estimate $(N-2)/\sum_j y_j^2$:

$$\widehat{\mu}_i = \left(1 - \frac{N-2}{\sum_j y_j^2}\right) y_i \quad \text{(James–Stein)}.$$

A quadratic $\log m \Rightarrow$ linear shrinkage $\Rightarrow$ James–Stein.

When is the "observable" route actually Bayes?

Because (T) needs only m, one may model m directly ("f-modeling") instead of F ("g-modeling").

- **Lindsey:** $\log m(y) = \beta_0 + \sum_{k=1}^{K} \beta_k y^k$, fit by Poisson regression.
- **Shape constraint:** $y \mapsto \mathbb{E}(\mu \mid y)$ is nondecreasing $\iff$ $c(y) = \frac{1}{2}y^2 + \log m(y)$ is convex.

A foundational question[1]

For which priors F can a given observable rule arise? A *polynomial* $\log m$ of degree ≥ 3 comes from *no* prior; degree 2 forces a Gaussian prior.

Takeaway to carry forward: the marginal m holds all the Bayesian content.

[1] self-plug

Just enough SDE

From ODE to SDE

You know the ODE, best read in increment form (Euler):

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x,t) \qquad \Longleftrightarrow \qquad x_{t+\Delta t} \approx x_t + f(x_t,t)\,\Delta t.$$

An SDE keeps the drift and adds a random kick at every step:

$$\mathrm{d}X_t = \underbrace{f(X_t,t)\,\mathrm{d}t}_{\text{drift = the ODE}} + \underbrace{g(X_t,t)\,\mathrm{d}B_t}_{\text{random kick}},$$

$$X_{t+\Delta t} \approx X_t + f(X_t,t)\,\Delta t + g(X_t,t)\,\sqrt{\Delta t}\;Z, \qquad Z \sim \mathcal{N}(0,1).$$

An SDE is an ODE with a wiggle. The one new fact: the kick scales as $\sqrt{\Delta t}$.

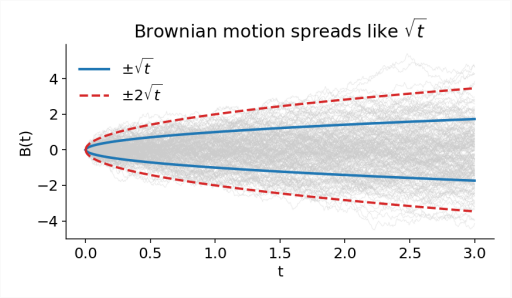
Brownian motion and the $\sqrt{\Delta t}$ rule

B_t (Wiener process) is the accumulated Gaussian kicks:

- $B_0 = 0$;
- independent increments;
- $B_t - B_s \sim \mathcal{N}(0,\, t - s)$.

Why $\sqrt{\Delta t}$? Independent kicks: *variances* add, not standard deviations. Over $[0, 1]$ with $n = 1/\Delta t$ steps, total variance $\propto n \cdot$ (per-step var). For a finite limit each kick needs variance $\propto \Delta t$, i.e. sd $\propto \sqrt{\Delta t}$.

Same $\sqrt{n}$ bookkeeping as the CLT.



SDE (contd.)

Simulate by Euler–Maruyama (the increment form):

$$X_{t+\Delta t} = X_t + f(X_t, t)\,\Delta t + g(X_t, t)\,\sqrt{\Delta t}\;Z.$$

You simulate it by Euler with $\sqrt{\Delta t}$ noise; there's a fancier chain rule called Itô's lemma that we won't need.

Itô's lemma

Since $\mathrm{d}B_t$ is $\sqrt{\Delta t}$-sized, $(\mathrm{d}B_t)^2$ is of order Δt: *not* negligible like $(\Delta t)^2$. Hence Itô's chain rule carries an extra second-derivative term.

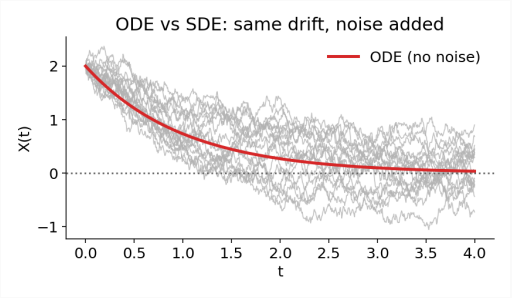
For *understanding* diffusion we only need the increment reading above.

Worked example: Ornstein–Uhlenbeck (continuous AR(1))

$$\mathrm{d}X_t = -\theta\,(X_t - \mu)\,\mathrm{d}t + \sigma\,\mathrm{d}B_t.$$

- Drift pulls toward μ (mean reversion).
- $\mathbb{E}(X_t) = \mu + (X_0 - \mu)e^{-\theta t}$: the ODE curve.
- Stationary law $\mathcal{N}\big(\mu,\ \sigma^2/(2\theta)\big)$.

This is the continuous-time analogue of AR(1).



red: ODE mean; grey: SDE paths

Two equivalent views of the same process

Particle view (SDE)

Follow one noisy path

$$\mathrm{d}X_t = f\,\mathrm{d}t + g\,\mathrm{d}B_t.$$

A different curve each run.

Density view (Fokker–Planck / PDE)

Follow the whole cloud $p(x, t)$:

$$\partial_t p = -\partial_x\big(f\,p\big) + \tfrac{1}{2}\,\partial_{xx}\big(g^2 p\big).$$

Deterministic evolution of the density.

Diffusion models live in both views: the forward SDE spreads p_{data} into a Gaussian: literally a diffusion of probability mass.

Diffusion models

The forward process

The simplest "variance-exploding" forward process just adds growing noise:

$$X_t = X_0 + \sigma_t Z, \qquad Z \sim \mathcal{N}(0, I), \qquad \Longrightarrow \qquad X_t \mid X_0 \sim \mathcal{N}(X_0,\ \sigma_t^2 I).$$

More generally it is an SDE $\mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}B_t$ with p_t the density of X_t, chosen so that X_T is (almost) pure noise.

Note: $X_t \mid X_0 \sim \mathcal{N}(X_0, \sigma_t^2)$ is a normal-means model at noise level σ_t, with p_{data} as the prior.

The reverse process needs only the score

Time reversal (Anderson)

The reverse-time process is again an SDE, and the only new ingredient is the score $\nabla \log p_t$:

$$\mathrm{d}X_t = \big[\, f(X_t, t) - g(t)^2 \, \nabla \log p_t(X_t) \,\big]\, \mathrm{d}t + g(t)\, \mathrm{d}\bar{B}_t,$$

run backward from $t = T$ to $t = 0$.

- Everything is known except $\nabla \log p_t$.
- Learn that one vector field, at every noise level, and you can generate.

The bridge: reverse diffusion is Tweedie at every level

For the forward model $X_t = X_0 + \sigma_t Z$, the best (posterior-mean) denoiser is, by Tweedie (T),

$$\boxed{D_t(x) := \mathbb{E}(X_0 \mid X_t = x) \;=\; x + \sigma_t^2 \, \nabla \log p_t(x).} \qquad \text{(D)}$$

Compare with the normal-means Tweedie formula:

$$\text{(T)} \quad \mathbb{E}(\mu \mid y) = y + \sigma^2 (\log m)'(y) \qquad \longleftrightarrow \qquad \text{(D)} \quad \mathbb{E}(X_0 \mid x) = x + \sigma_t^2 \, \nabla \log p_t(x).$$

They are the same identity. Diffusion stitches these denoisers together over a ladder of noise levels σ_t via the reverse SDE.

How the score is learned (denoising score matching)

We cannot evaluate $\nabla \log p_t$ directly. But fitting a network $s_\theta(x, t)$ to the *tractable* conditional score of the forward kernel,

$$\min_\theta \ \mathbb{E}\big\| s_\theta(X_t, t) - \nabla_x \log q(X_t \mid X_0) \big\|^2, \qquad \nabla_x \log q(x \mid X_0) = -\frac{x - X_0}{\sigma_t^2},$$

has the same minimiser as matching the intractable marginal score $\nabla \log p_t$.

Equivalent parameterisations (all the same object)

noise prediction ε_θ, clean prediction x_θ, and the score s_θ are reparameterisations of one another; the optimum is $x_\theta^\star(x_t, t) = \mathbb{E}(X_0 \mid X_t = x_t)$: the denoiser again.

Sampling: the reverse step, in words

Discretise the reverse SDE (variance-exploding, $g = 1$):

$$x \leftarrow x + \underbrace{\nabla \log p_t(x)}_{\text{move toward data}} \; h \; + \; \underbrace{\sqrt{h}\, Z}_{\text{fresh noise}}, \qquad \text{stepping } t \text{ downward.}$$

- Each step nudges the sample *along the score* (toward higher density) and injects a little noise.
- Start from noise at $t = T$; end near a data sample at $t \approx 0$.
- Replacing $\nabla \log p_t$ by a trained s_θ is the *only* difference from a real model.

A baby/toy example

Images are the 1-D mixture case in disguise

Treat the data as a tiny set of stored images $X_1, \ldots, X_K$ (point masses). Adding noise gives a Gaussian mixture,

$$p_t = \frac{1}{K} \sum_{k=1}^{K} \mathcal{N}(X_k,\, (v_0 + t) I) \qquad (\sigma_t^2 = t,\ v_0 \text{ a small floor}),$$

whose exact score is a responsibility-weighted average: *now in pixel space*. By Tweedie (D),

$$D_t(x) = \sum_k r_k(x)\, X_k \qquad \text{(weighted average of the stored images)}.$$

Noising one image and reversing it is the $K = 1$ case: score $= -(x - X_1)/(v_0 + t)$. No neural network needed.

Forward then reverse, on an actual image

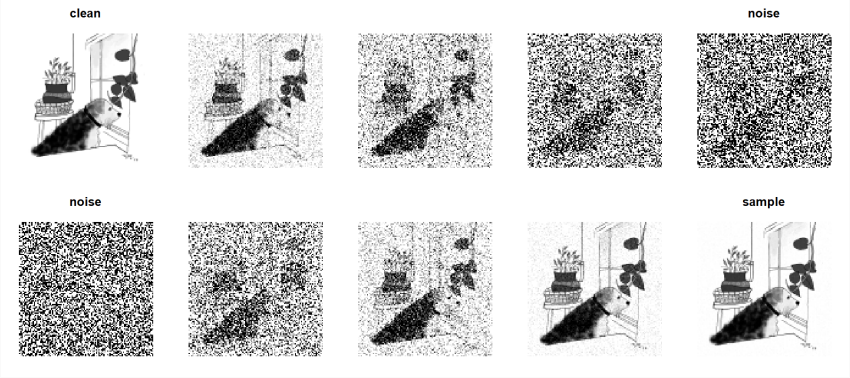


Figure 1: Top: forward: clean image → noise. **Bottom:** reverse: noise → recovered sample, driven by the exact score.

- Same code as the 1-D toy, scalar mean → image.
- $K = 1$: the model can only reproduce its one training image (memorisation); $K > 1$ gives real generation.

Tail behaviour: a Tauberian theorem

Normal variance mixtures

Let $X \mid Z \sim \mathcal{N}(0, Z)$ with mixing variable $Z \sim \pi$ on $(0, \infty)$; equivalently $X = \sqrt{Z}\, G$, $G \sim \mathcal{N}(0,1)$:

$$p(x) = \int_0^\infty \frac{1}{\sqrt{2\pi z}}\, e^{-x^2/(2z)}\, \pi(z)\, \mathrm{d}z.$$

Every global–local shrinkage prior (horseshoe, Student-t, Laplace) has this form.

Question. How does the tail of the mixing law π determine the tail of p?

Barndorff-Nielsen's transfer theorem

Suppose the mixing density has the tail

$$\pi(z) \sim z^{\lambda-1}\, e^{-\psi_+ z}\, L(z), \qquad z \to \infty,$$

with L slowly varying and $\psi_+ \geq 0$ the abscissa of the mgf of π. Then the tail transfers through the Gaussian kernel:

$$p(x) \sim C\, |x|^{2\lambda-1}\, L(x^2), \qquad \psi_+ = 0 \;\; \text{(heavy / polynomial)},$$

$$p(x) \sim C'\, |x|^{\lambda-1}\, e^{-\sqrt{2\psi_+}\,|x|}\, L(|x|), \quad \psi_+ > 0 \;\; \text{(exponential)}.$$

The single number ψ_+ decides heavy (polynomial) vs. light (exponential).

From marginal tail to score to denoiser

Differentiate the two tails to get the marginal score:

$$(\log p)'(x) \sim \begin{cases} \dfrac{2\lambda - 1}{x} \longrightarrow 0, & \psi_+ = 0, \\ -\sqrt{2\psi_+}\ \operatorname{sgn}(x) + \dfrac{\lambda - 1}{x}, & \psi_+ > 0. \end{cases}$$

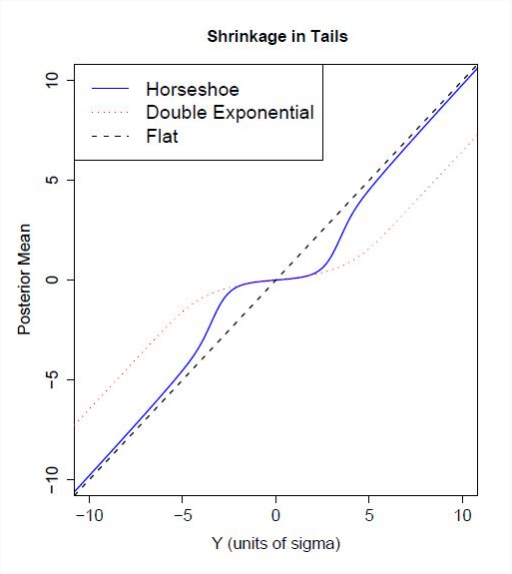
Now feed this into Tweedie, $D_t(x) = x + \sigma_t^2 (\log p_t)'(x)$:

- heavy: the correction *vanishes* in the tail;
- exponential: the correction *saturates* at a constant.

The denoiser trichotomy

Data tail	Denoiser D_t
Heavy ($\psi_+ = 0$) Student-t, horseshoe	$D_t(x) \to x$: redescending
Exponential ($\psi_+ > 0$) Laplace	$D_t(x) - x \to$ const: saturating
Gaussian (degenerate π)	$D_t(x) = \frac{\tau^2}{\tau^2+\sigma_t^2} x$: linear

Heavy tails $\Rightarrow$ a denoiser that *stops correcting* far out: it treats a large input as genuine signal (horseshoe tail-robustness).



Why this matters for diffusion

If the data are a scale mixture $X_0 = \sqrt{V}\, G$ with $V \sim F$, then adding noise $\sigma_t^2 = t$ gives

$$p_t = \int_0^\infty \mathcal{N}\big(0,\, (v+t)I\big)\, \mathrm{d}F(v) \quad = \quad \text{a normal variance mixture with mixing } V + t.$$

Key point

Translating the mixing law by t changes neither ψ_+ nor λ. So the tail *regime* (heavy / exponential / Gaussian) of the data persists at every noise level, and fixes the shape of the optimal denoiser all along the reverse trajectory.

The tail of p_{data} controls the tail behaviour of the diffusion denoiser.

Summary

Three things to take away

1. **Tweedie's formula** writes the Bayes rule as "unbiased estimate + a correction that depends only on the marginal m."
2. **A diffusion model** applies exactly this correction at every noise level: $D_t(x) = x + \sigma_t^2 \nabla \log p_t(x)$, stitched by the reverse SDE. Score matching = empirical Bayes at every scale.
3. **The Barndorff-Nielsen transfer theorem** shows the tail of the data, encoded in the mixing law, decides whether the optimal denoiser is redescending, saturating, or linear.

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